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Project #157: Improving Central Line Necessity Documentation: A Quality Improvement Project

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Abstract

Introduction: Central line associated blood stream infections (CLABSI) are a preventable infection that can cause severe illness, increased length of stay, and increase healthcare costs. ^{3,4,5}

Methods: Literature reviewed and the Plan-Do-Check-Act (PDCA) was used to develop a plan to meet the set hospital benchmark for documentation and increase awareness regarding CLABSI prevention

Results: Hospital goal for central line necessity documentation goal of 70% or greater was achieved. H4 and I4 average rate of documentation of 80% for H4 and 78% for I4 (since March 2024); There were no CLABSI events on I4 and two CLABSI events on H4

Conclusions: Using a multi-pronged approach and the framework of the PDCA cycle the team was able to meet the set documentation and impact CLABSI rates

Introduction

- The CDC *Checklist for Prevention of Central Line Associated Bloodstream Infection* provides guidance to healthcare providers; including necessity documentation & prompt line removal
- As part of CLABSI reduction initiatives, HFH set goals for inpatient units for central line necessity documentation
 - Baseline data showed our units were not at goal
- The goal of this project was to implement daily central line necessity documentation into our daily practice on the acute care surgery (ACS) service using the PDCA cycle

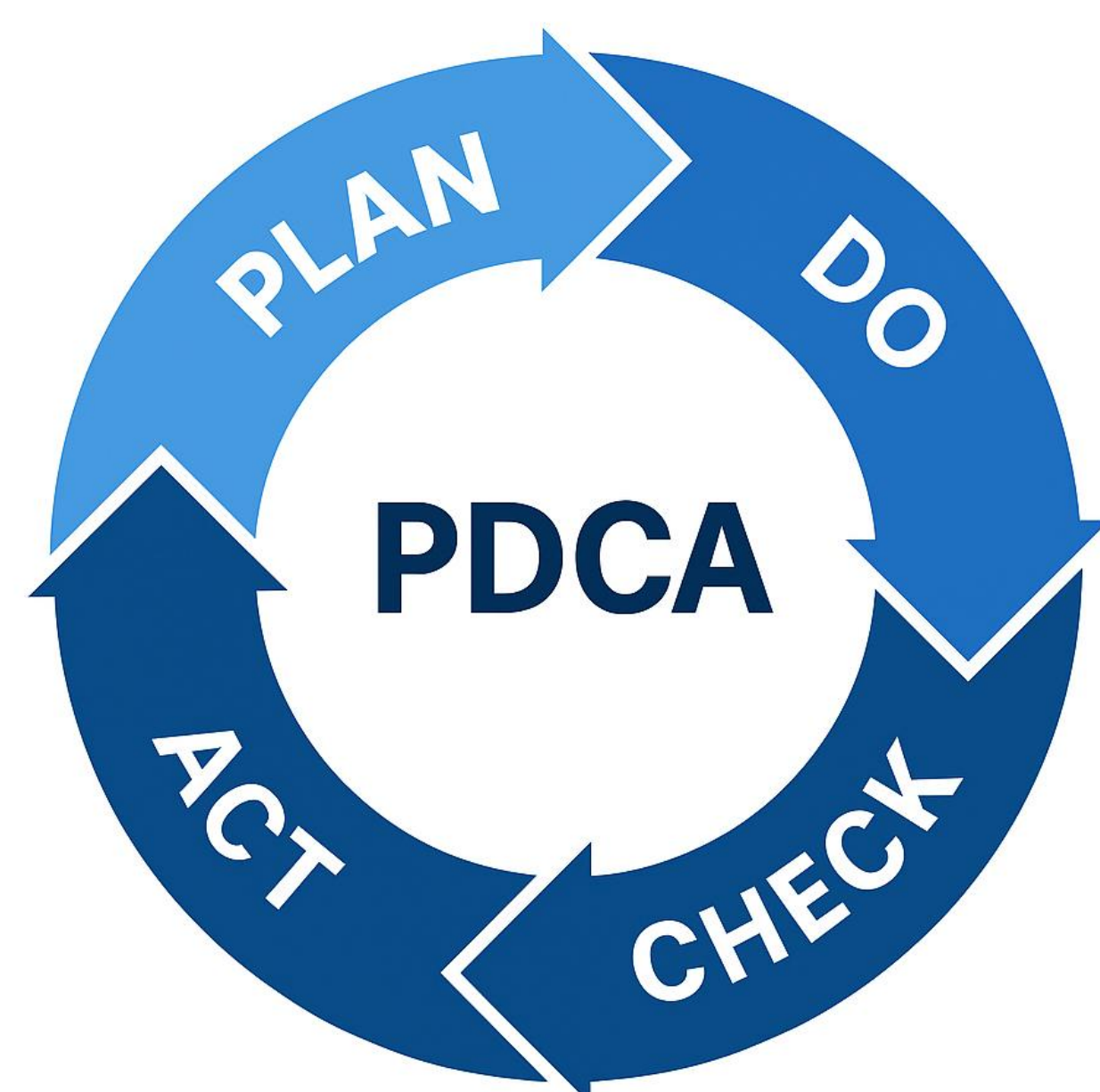


Image 1: PDCA Cycle

Methods

Current practice of central line necessity documentation was reviewed by ACS NP leadership

- Literature reviewed
- Gaps in practice identified
- Interventions chosen based on noted gaps &

The following interventions were implemented over a 16-month period

- Documentation Education** for all team members
 - Adding column to patient list (visual reminder)
 - Required elements of documentation
- Data sharing:** Monthly open sharing of documentation of line necessity using HFH PowerBI report
 - Including gaps and successes at team staff meeting
 - Addressing questions regarding report & data
- A set reminder** to cue documentation of central line necessity on team calendar for days commonly missed
- Collaboration** with APPs on other surgical services regarding documentation
- CLABSI Prevention Education:** sharing of current literature, articles and education session by infection control
- Collaboration** with nursing staff for timely line removal

Results

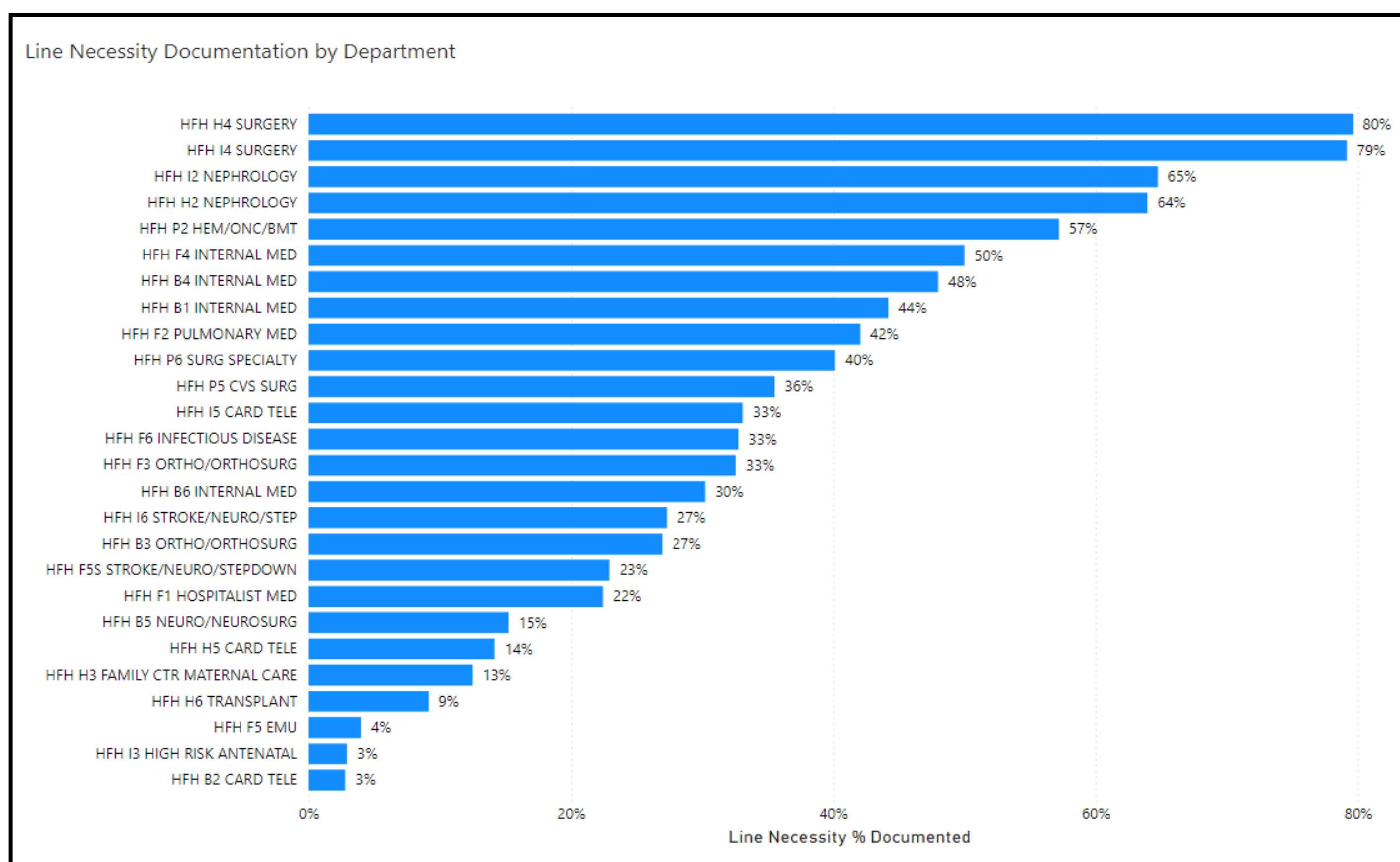


Image 2: Central Line Necessity Documentation HFH GPU March 20204-September 4, 2024

Discussion

Summary

- A cost-effective faceted approach led to achievement of set goal; however, missed opportunities can still exist
- Gaps are evaluated monthly for opportunities for improvement
- Practice change was aided by involving staff in data driven conversations & removing obstacles

Limitations

- Simultaneous efforts for CLABSI reduction by nursing, CHG bathing & Nasal Sanitization may have also impacted CLABSI events
- Overcoming resistance to change & sustaining change takes time and leadership
- Implemented interventions were effective for one service line; may not be translatable to others

Conclusion

- Continued efforts are needed to sustain daily documentation
- Next steps include:
 - timely discontinuation of unnecessary central lines
 - Reducing time till central line removed in collaboration with the unit nursing staff.



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